



LECTURE #15

Land Use Data Analysis in QGIS

OBJECTIVE

ANALYSE URBAN LAND USE DATA IN QGIS BY NEIGHBORHOOD USING VECTOR TOOLS AND BASIC STATISTICS.

STEPS

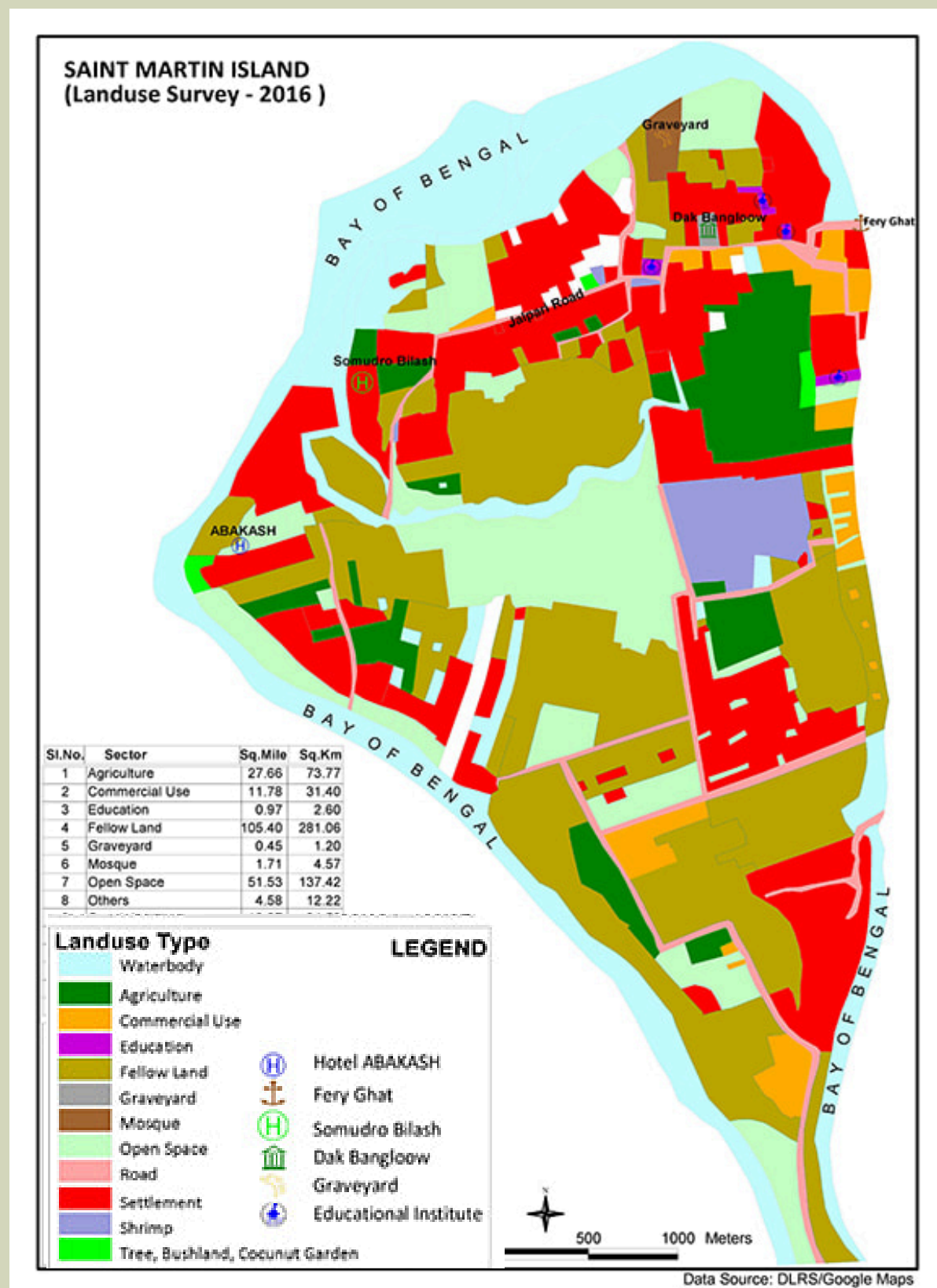
- INTRODUCTION TO LAND USE DATA AND WHY IT IS IMPORTANT.
- ADDING AND PREPARING LAYERS (CLIPPING LAND USE TO FREIBURG'S BORDER AND SETTING CATEGORIZED SYMBOLOGY).
- FORMULATING SPATIAL QUESTIONS ABOUT LAND USE PATTERNS BY NEIGHBORHOOD.
- APPLYING QGIS VECTOR TOOLS (INTERSECTION, FIELD CALCULATOR, FILTER/SELECT BY EXPRESSION, STATISTICS BY CATEGORIES, JOIN ATTRIBUTES BY LOCATION).
- INTERPRETING RESULTS (MOST FREQUENT LAND USE, RESIDENTIAL AREAS, GREEN/NATURAL LAND USE, COMMERCIAL/RETAIL AREAS, LAND USE DIVERSITY, PARKS AND SOCIAL GREEN SPACES).

LAND USE DATA

A LAND USE SHAPEFILE IS USUALLY CREATED FROM SATELLITE IMAGE CLASSIFICATION, AND LATER REFINED USING TOPOGRAPHIC SURVEYS, CADASTRAL INFORMATION, OR OFFICIAL GOVERNMENT DATASETS. IT REPRESENTS HOW THE LAND IS BEING USED: RESIDENTIAL AREAS, COMMERCIAL ZONES, INDUSTRIAL SITES, GREEN SPACES, FORESTS, AGRICULTURE, AND SO ON.

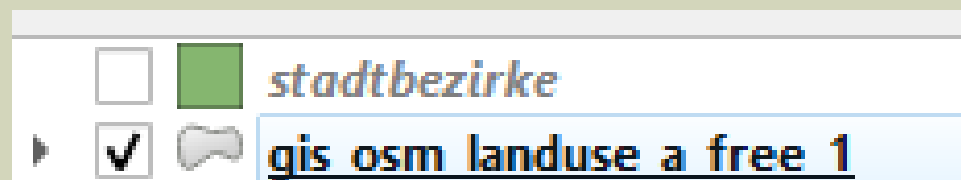
2) WHY LAND USE DATA IS IMPORTANT

IT IS IMPORTANT FOR URBAN PLANNING, ENVIRONMENTAL MANAGEMENT, AND DECISION-MAKING. IT HELPS CITIES UNDERSTAND HOW SPACE IS ORGANIZED (WHERE GREEN AREAS ARE LACKING, AND HOW URBAN EXPANSION IS HAPPENING...). WHEN COMBINED WITH ADMINISTRATIVE UNITS SUCH AS NEIGHBORHOODS, WE CAN DESCRIBE SPATIAL PATTERNS AND COMPARE SOCIO-ENVIRONMENTAL DIFFERENCES WITHIN THE CITY.



LET'S ANALYSE FREIBURG'S LAND USE BY NEIGHBORHOOD

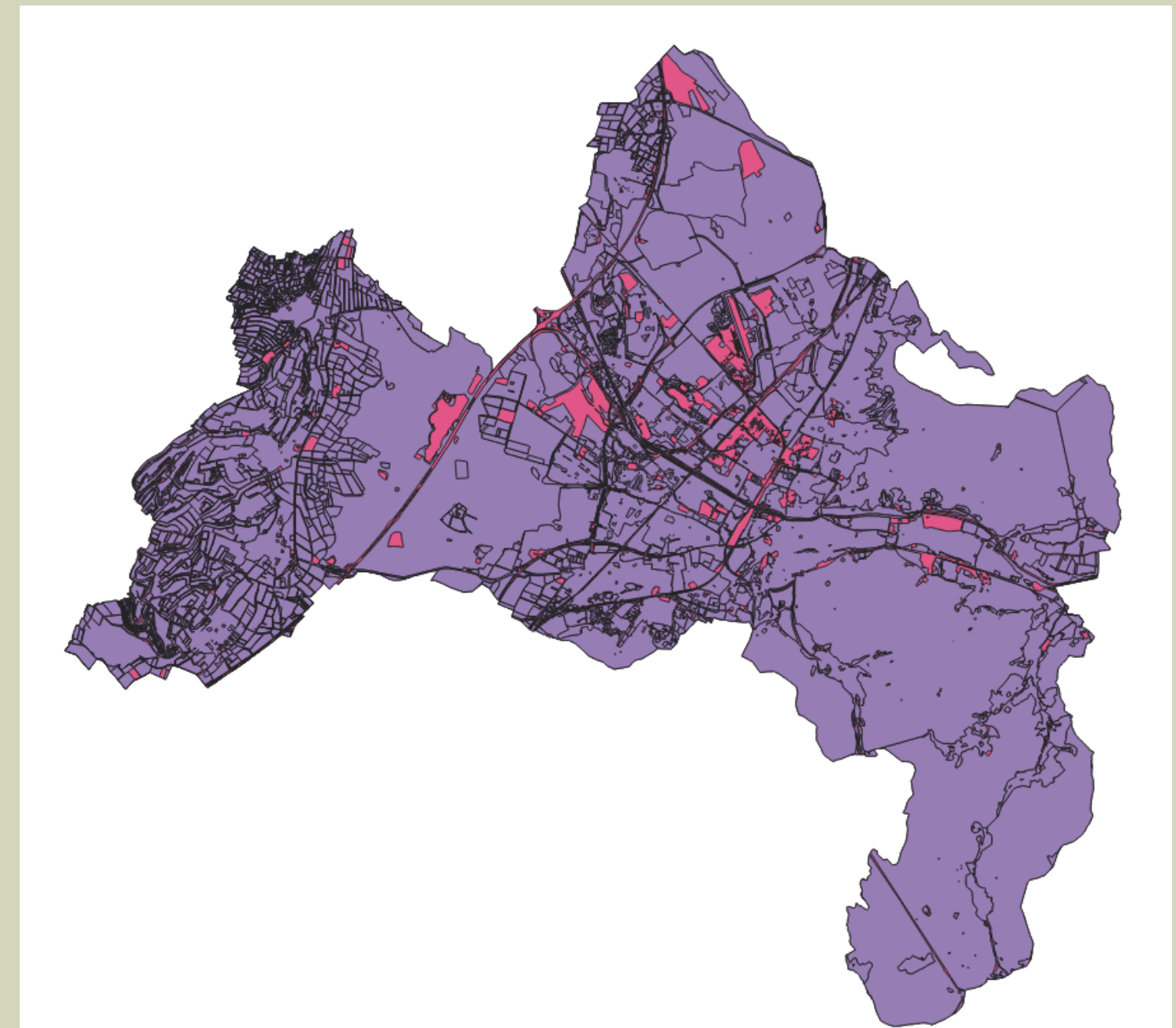
ADD LAYERS:



LAND USE FROM THIS DATASET:
FREIBURG-REGBEZ-251026-FREE.SHP

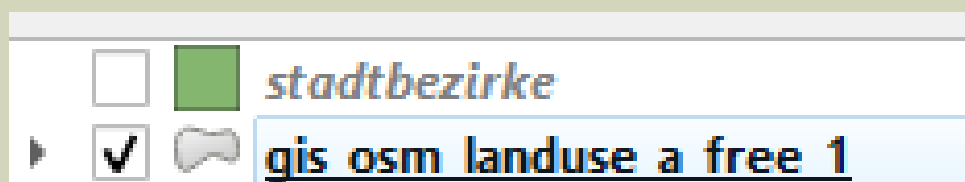
CLIP LAND USE SHAPEFILE TO FREIBURG'S BORDER.

- REMEMBER TO KEEP BOTH LAYERS IN THE SAME COORDINATE SYSTEM BEFORE CLIPPING.



LET'S ANALYSE FREIBURG'S LAND USE BY NEIGHBORHOOD

ADD LAYERS:



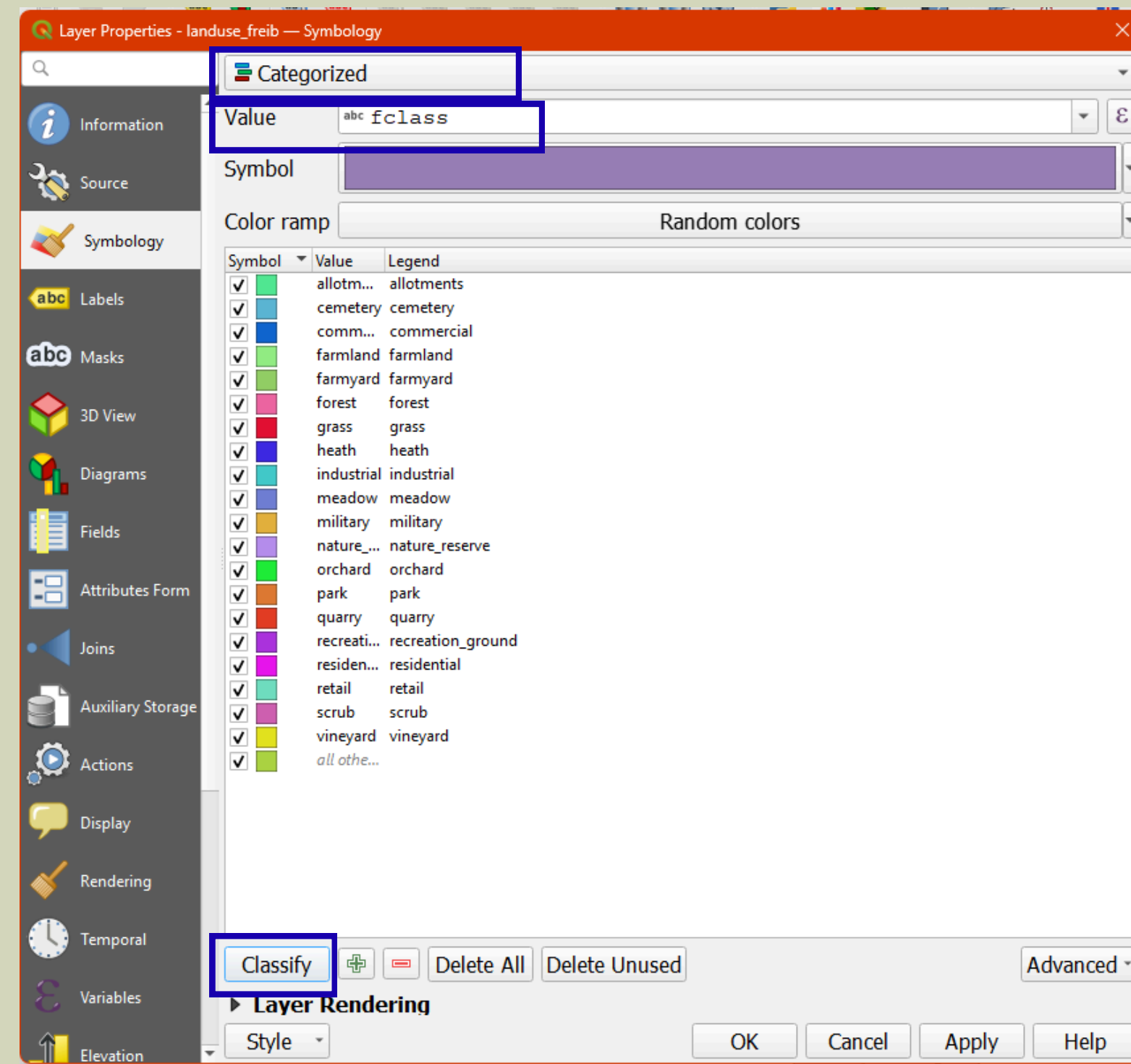
LAND USE FROM THIS DATASET:
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CLIP LAND USE SHAPEFILE TO FREIBURG'S BORDER.

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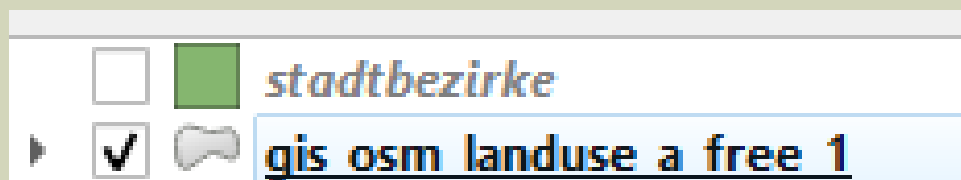
IN THE LAYER'S PROPERTIES, YOU CAN CHANGE THE SYMBOLOGY TO "CATEGORIZED" TO COLOUR THE LAND USE CLASSES.

- VALUE: FCLASS
- CLICK 'CLASSIFY' TO CREATE SEE THE LIST.



LET'S ANALYSE FREIBURG'S LAND USE BY NEIGHBORHOOD

ADD LAYERS:



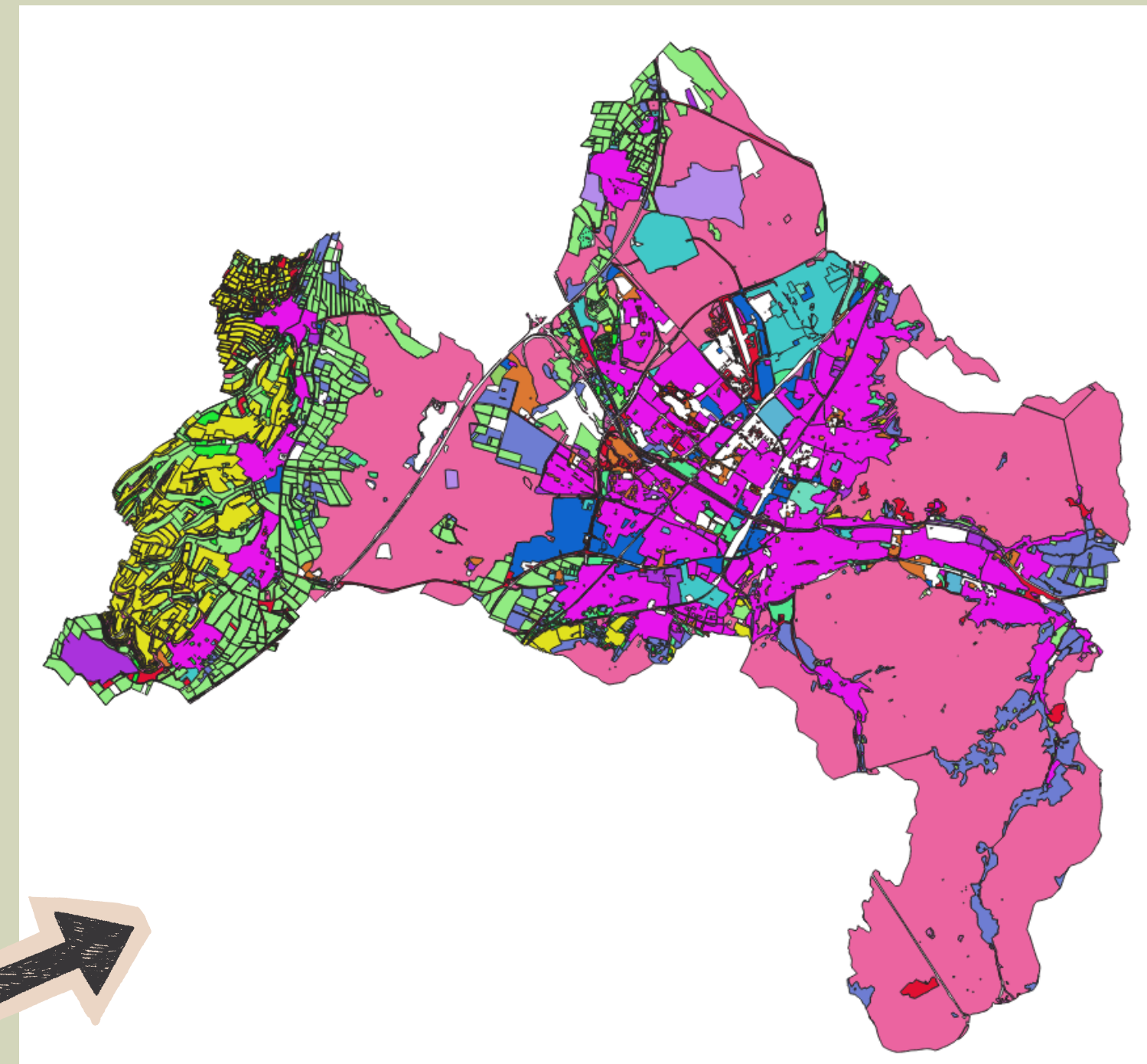
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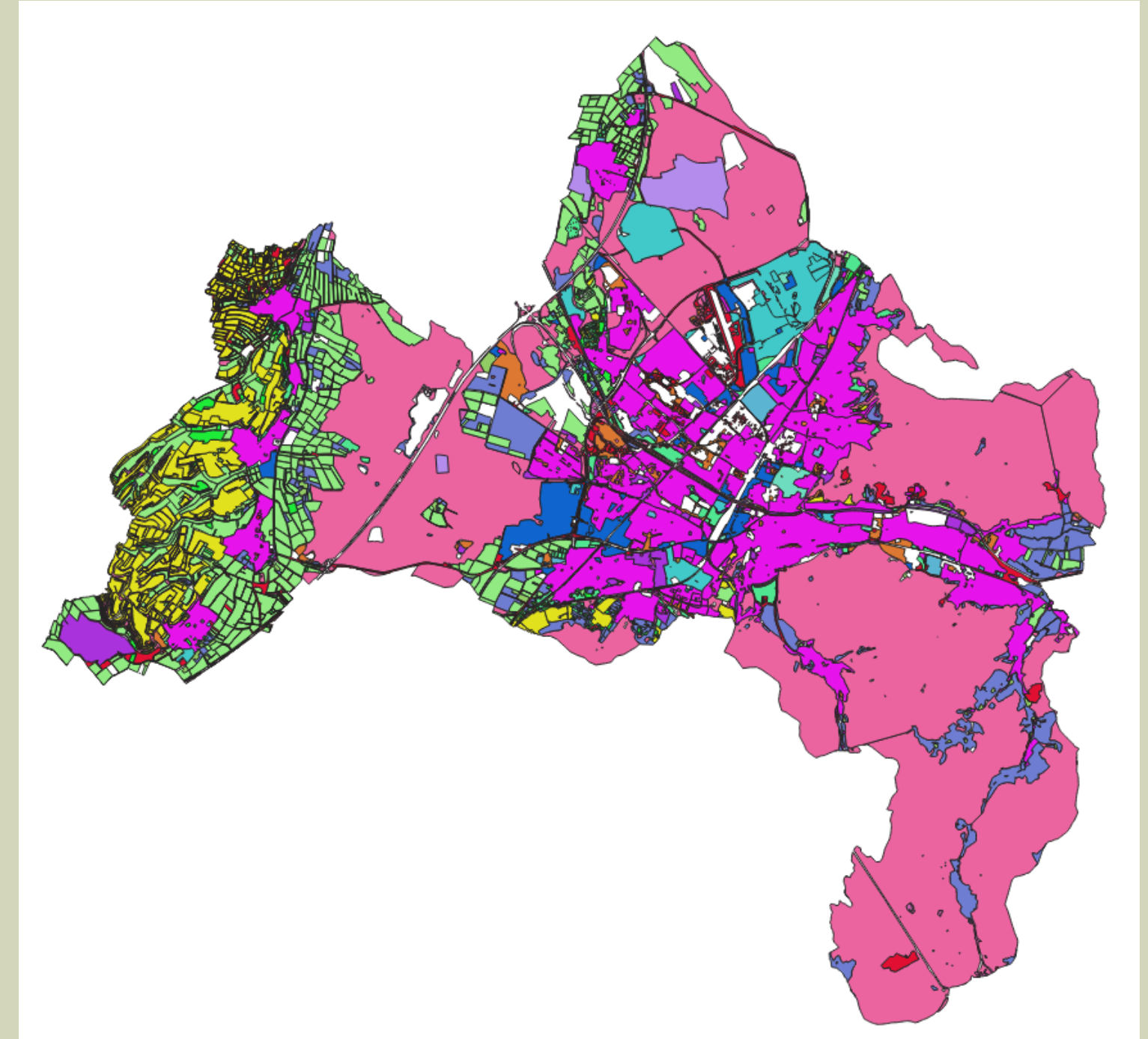
- VALUE: FCLASS
- CLICK 'CLASSIFY' TO CREATE SEE THE LIST.



LET'S ANALYSE FREIBURG'S LAND USE BY NEIGHBORHOOD

SOME SPATIAL QUESTIONS:

- 1) WHAT IS THE MOST FREQUENT LANDUSE IN FREIBURG?
- 2) WHICH NEIGHBORHOOD HAS THE LARGEST RESIDENTIAL AREA?
- 3) WHICH NEIGHBORHOODS HAVE THE HIGHEST AMOUNT OF GREEN AND NATURAL LAND USE?
- 4) WHERE ARE COMMERCIAL AND RETAIL AREAS CONCENTRATED?
- 5) WHICH NEIGHBORHOODS HAVE THE HIGHEST LAND USE DIVERSITY?
- 6) WHICH NEIGHBORHOODS HAVE THE HIGHEST NUMBER OF PARKS AND SOCIAL GREEN SPACES?



Q1: What is the most frequent landuse in Freiburg?

06

1) GO TO:

PROCESSING TOOLBOX → VECTOR ANALYSIS → BASIC STATISTICS FOR FIELDS

INPUT LAYER: LANDUSE

FIELD TO CALCULATE STATISTICS ON: FCLASS

RUN

2) OPEN THE OUTPUT'S ATTRIBUTE TABLE.

Statistics — Features Total: 1, Filtered: 1, Selected: 0

	count	unique	empty	filled	min	max	min_length	max_length	mean_length	minority	majority
1	5219	20	0	5219	allotments	vineyard	4	17	6,74190457942...	military	grass

Q2: Which neighborhood has the largest residential area?

07

1) INTERSECT NEIGHBORHOODS WITH THE LAND USE LAYER

QGIS → VECTOR → GEOPROCESSING TOOLS → INTERSECTION

IT GIVES YOU ONLY THE PARTS OF LAND USE INSIDE EACH NEIGHBORHOOD.

INPUT LAYER: NEIGHBORHOODS

OVERLAY LAYER: LAND USE

RUN

YOU WILL GET A NEW LAYER WHERE EACH POLYGON HAS:

THE NEIGHBORHOOD ID/NAME

THE LAND USE CLASS (E.G., RESIDENTIAL)

LAND USE GEOMETRY IS CLIPPED TO THE NEIGHBORHOOD BOUNDARY.

2) CALCULATE THE AREA OF EACH PIECE OF LAND USE

ADD AN AREA FIELD:

ATTRIBUTE TABLE → FIELD CALCULATOR → CREATE NEW FIELD

FIELD NAME: AREA_HA

EXPRESSION: \$AREA / 10000

OUTPUT: DECIMAL NUMBER

NOW EACH POLYGON (EACH LAND USE FRAGMENT) HAS ITS AREA IN HECTARES.

Q2: Which neighborhood has the largest residential area?

3) FILTER THE LAYER TO RESIDENTIAL ONLY.

RIGHT BUTTON ON THE LAYER → FILTER →
EXPRESSION: "FCLASS" LIKE 'RESIDENTIAL'

4) SUMMARIZE RESIDENTIAL AREA BY NEIGHBORHOOD

VECTOR → ANALYSIS TOOLS → STATISTICS BY CATEGORIES

CATEGORIZED BY: NEIGHBORHOOD NAME

FIELD TO CALCULATE SUM: AREA_HA

THE NEIGHBORHOOD WITH THE HIGHEST SUM IS THE
ANSWER.



Statistics by category — Features Total: 42, Filtered: 42, Selected: 0

	name	count	unique	min	max	range	sum	mean
1	St. Georgen-No...	4	4	0,24	129,364	129,124	131,133	32,78
2	Zähringen	9	9	0,102	59,379	59,277	111,901	12,433444444
3	Littenweiler	7	7	0,102	80,137	80,035	111,412999999...	15,916142857
4	Unterwihre-Süd	9	9	0,004	77,48	77,476	100,242	11,1
5	Herdern-Nord	3	3	0,42	83,789	83,369	87,194	29,064666666
6	Herdern-Süd	5	5	0,018	80,735	80,717	81,726999999...	16,345399999
7	Waldsee	7	7	0,018	74,763	74,745	81,529	11,6
8	Haslach-Garten...	3	3	0,003	79,04	79,037	79,124000000...	26,374666666
9	Hochdorf	2	2	7,229	70,44	63,211	77,669	38,83
10	Weingarten	6	6	0,35	40,519	40,169	71,611999999...	11,935333333
11	Opfingen	5	5	0,009	55,854	55,845	67,448000000...	13,489600000
12	Mooswald-West	2	2	0,437	65,021	64,584	65,458	32,7
13	Landwasser	2	2	0,089	63,565	63,476	63,653999999...	31,826999999
14	Haslach-Egerten	8	8	0,565	41,715	41,150000000...	62,058	7,757
15	Tiengen	2	2	0,336	61,708	61,372	62,044	31,0
16	Betzenhausen-...	6	6	1,345	30,556	29,211000000...	61,559	10,259833333
17	Rieselfeld	3	3	0,116	55,909	55,793	60,385	20,128333333
18	Neuburg	3	3	0,22	53,149	52,929	57,138	19,0

Show All Features

Q3: Which neighborhoods have the highest amount of green and natural land us?

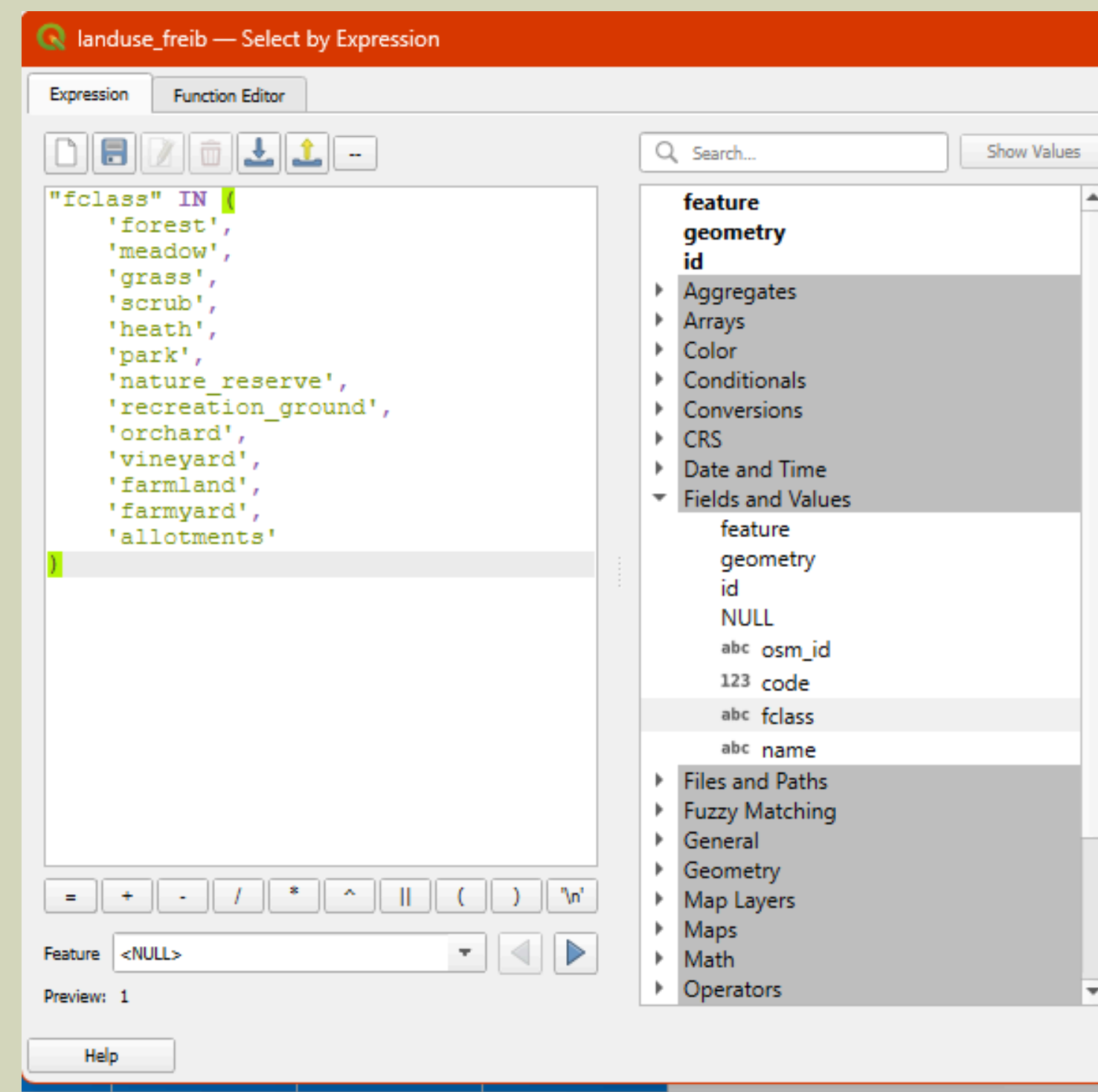
1) FILTER THE NATURAL LAND USE CLASSES

OPEN THE ATTRIBUTE TABLE → SELECT BY EXPRESSION:
EXPRESSION: "LANDUSE" IN (

'FOREST',
'MEADOW',
'GRASS',
'SCRUB',
'HEATH',
'PARK',
'NATURE_RESERVE',
'RECREATION_GROUND',
'ORCHARD',
'VINEYARD',
'FARMLAND',
'FARMYARD',
'ALLOTMENTS'

)

THIS KEEPS ONLY THE NATURAL/GREEN CATEGORIES.



Q3: Which neighborhoods have the highest amount of green and natural land us?

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CONTINUE FOLLOWING THE PROCESS OF QUESTION 1:

2) INTERSERCT;

3) CALCULATE AREA

3) RUN "STATISTICS BY CATEGORIES"

4) INTERPRET THE RESULTS

SORT THE COLUMN SUM FROM LARGEST TO SMALLEST.
THE NEIGHBORHOODS AT THE TOP ARE THE ONES WITH:
THE HIGHEST AMOUNT OF GREEN AND NATURAL LAND USE.

Q4: Where are commercial and retail areas concentrated?

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1) FILTER THE COMMERCIAL AND RETAIL CLASSES

**OPEN ATTRIBUTE TABLE → SELECT BY EXPRESSION:
"FCLASS" IN ('COMMERCIAL', 'RETAIL')**

**THIS WILL KEEP ONLY THE POLYGONS CLASSIFIED AS
COMMERCIAL OR RETAIL.**

**CONTINUE FOLLOWING THE PROCESS OF QUESTION 1 AND
2:**

2) INTERSECT;

3) CALCULATE AREA

3) RUN "STATISTICS BY CATEGORIES"

4) INTERPRET THE RESULTS

**SORT THE COLUMN SUM FROM LARGEST TO SMALLEST.
THE NEIGHBORHOODS WITH THE HIGHEST SUM VALUES ARE
THE ONES WHERE COMMERCIAL AND RETAIL LAND USES
ARE MOST CONCENTRATED.**

Q5: Which neighborhoods have the highest land use diversity?

1) INTERSECT NEIGHBORHOODS WITH THE LAND USE LAYER (WITHOUT FILTER)

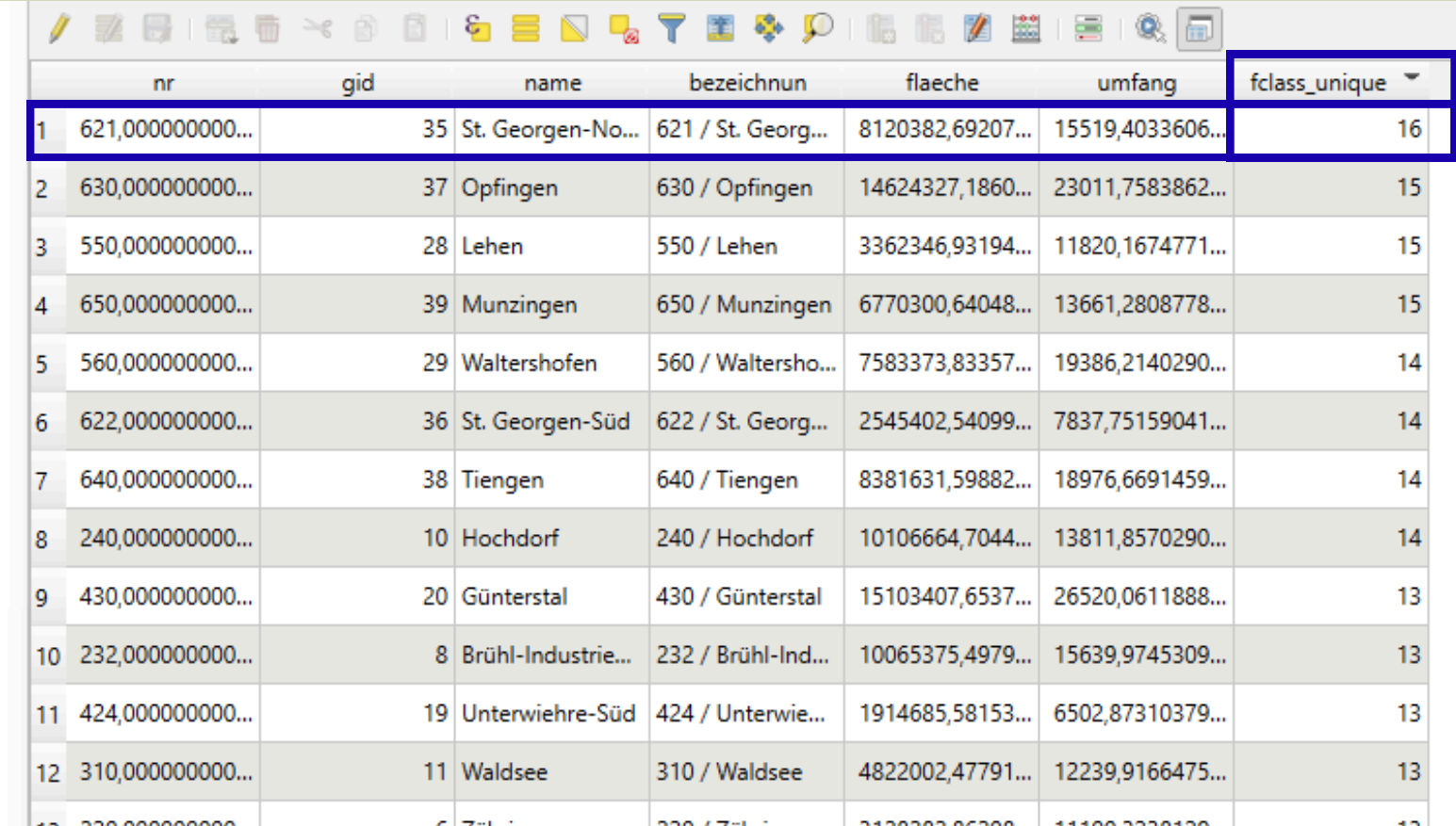
QGIS → VECTOR → GEOPROCESSING TOOLS → INTERSECTION

2) OPEN “JOIN ATTRIBUTES BY LOCATION (SUMMARY)”
VECTOR → DATA MANAGEMENT TOOLS → JOIN ATTRIBUTES BY LOCATION (SUMMARY)

- TARGET LAYER: NEIGHBORHOODS.
- JOIN LAYER: INTERSECTION LAYER.
- SPATIAL PREDICATE: SELECT INTERSECTS.
- FIELDS TO SUMMARISE: SELECT FCLASS.
- SUMMARIES TO CALCULATE: CHECK **UNIQUE**.

RUN THE TOOL TO CREATE A NEW NEIGHBORHOOD LAYER WITH A FIELD SHOWING THE NUMBER OF DISTINCT LAND-USE CLASSES PER NEIGHBORHOOD.

CHECK THIS FIELDS IN THE ATTRIBUTE TABLE.



	nr	gid	name	bezeichnung	flaeche	umfang	fclass_unique
1	621,000000000...	35	St. Georgen-No...	621 / St. Georg...	8120382,69207...	15519,4033606...	16
2	630,000000000...	37	Opfingen	630 / Opfingen	14624327,1860...	23011,7583862...	15
3	550,000000000...	28	Lehen	550 / Lehen	3362346,93194...	11820,1674771...	15
4	650,000000000...	39	Munzingen	650 / Munzingen	6770300,64048...	13661,2808778...	15
5	560,000000000...	29	Waltershofen	560 / Walters...	7583373,83357...	19386,2140290...	14
6	622,000000000...	36	St. Georgen-Süd	622 / St. Georg...	2545402,54099...	7837,75159041...	14
7	640,000000000...	38	Tiengen	640 / Tiengen	8381631,59882...	18976,6691459...	14
8	240,000000000...	10	Hochdorf	240 / Hochdorf	10106664,7044...	13811,8570290...	14
9	430,000000000...	20	Günterstal	430 / Günterstal	15103407,6537...	26520,0611888...	13
10	232,000000000...	8	Brühl-Industrie...	232 / Brühl-Ind...	10065375,4979...	15639,9745309...	13
11	424,000000000...	19	Unterwihre-Süd	424 / Unterwie...	1914685,58153...	6502,87310379...	13
12	310,000000000...	11	Waldsee	310 / Waldsee	4822002,47791...	12239,9166475...	13
13	220,000000000...	6	Zähringen	220 / Zähringen	312822,86208...	11100,3238129...	13

Q6: Which neighborhoods have the highest number of parks and social green spaces?

1) INTERSECT NEIGHBORHOODS WITH THE LAND USE LAYER (WITHOUT FILTER)

QGIS → VECTOR → GEOPROCESSING TOOLS → INTERSECTION

2) FILTER "FCLASS" IN ('PARK', 'RECREATION_GROUND')

3) OPEN "JOIN ATTRIBUTES BY LOCATION (SUMMARY)"
VECTOR → DATA MANAGEMENT TOOLS → JOIN ATTRIBUTES BY LOCATION (SUMMARY)

- TARGET LAYER: NEIGHBORHOODS.
- JOIN LAYER: INTERSECTION LAYER.
- SPATIAL PREDICATE: SELECT INTERSECTS.
- FIELDS TO SUMMARISE: SELECT FCLASS.
- SUMMARIES TO CALCULATE: CHECK COUNT.

CHECK THIS FIELDS IN THE ATTRIBUTE TABLE.

Joined layer — Features Total: 42, Filtered: 42, Selected: 0

	nr	gid	name	bezeichnung	flaeche	umfang	fclass_count
1	660,000000000...	40	Weingarten	660 / Weingarten	1630506,47384...	5540,20133940...	28
2	310,000000000...	11	Waldsee	310 / Waldsee	4822002,47791...	12239,9166475...	23
3	531,000000000...	25	Betzenhausen-...	531 / Betzenha...	1364642,80644...	5938,33571059...	21
4	410,000000000...	15	Oberau	410 / Oberau	1282206,52422...	5831,18022193...	18
5	640,000000000...	38	Tiengen	640 / Tiengen	8381631,59882...	18976,6691459...	16
6	112,000000000...	2	Altstadt-Ring	112 / Altstadt-R...	617913,651668...	4216,49596228...	15
7	512,000000000...	21	Stühlinger-Esch...	512 / Stühlinge...	901367,916191...	4086,20893256...	14
8	532,000000000...	26	Alt-Betzenhausen	532 / Alt-Betze...	798023,793315...	5021,52731244...	13
9	513,000000000...	22	Alt-Stühlinger	513 / Alt-Stühli...	837999,297458...	3986,36698807...	12
10	120,000000000...	3	Neuburg	120 / Neuburg	1636432,60775...	7141,30579814...	12
11	330,000000000...	13	Ebnet	330 / Ebnet	6868398,30259...	12621,7484235...	12
12	521,000000000...	23	Mooswald-West	521 / Mooswal...	968797,148604...	4031,36407260...	11

Show All Features